EXTENSION BY ZERO OF FINITE CONSTANT SHEAVES IS CONSTRUCTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all three parts of the statement.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Representable sheaf h_U and extension by zero j_! of constant sheaves. Same chapter, the sheaf represented by an etale scheme and the functor j_! applied to constant sheaves. Supplies the sheaves whose constructibility is asserted.
- Constructibility by a partition into constructible locally closed strata. Same chapter, the characterisation of constructibility through a partition and the result that an etale morphism of quasi-compact quasi-separated schemes is finite etale over the strata of a suitable partition. Supplies the partition with respect to which the sheaves are shown constructible.

Source reference. Original lemma 03S0: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12035–12035. Navigation pointer invoked at proof line 12199; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03RV: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9994–9994. Navigation pointer invoked at proof line 12202; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 03S6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11493–11493. Navigation pointer invoked at proof line 12209; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03S7: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11571–11571. Navigation pointer invoked at proof line 12210; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 095B: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9973–9973. Navigation pointer invoked at proof line 12212; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $j : U \to X$ be an étale morphism of quasi-compact and quasi-separated schemes.*

1. *The sheaf h_U is a constructible sheaf of sets.*
2. *The sheaf $j_!\underline{M}$ is a constructible abelian sheaf for a finite abelian group M.*
3. *If Λ is a Noetherian ring and M is a finite Λ-module, then $j_!\underline{M}$ is a constructible sheaf of Λ-modules on $X_{étale}$.*

Proof. By Lemma 03S0 there is a partition $\coprod_i X_i$ such that $\pi_i : j^{-1}(X_i) \to X_i$ is finite étale. The restriction of h_U to X_i is $h_{j^{-1}(X_i)}$ which is finite locally constant by Lemma 03RV. For cases (2) and (3) we note that

$$j_!(\underline{M})|_{X_i} = \pi_{i!}(\underline{M}) = \pi_{i*}(\underline{M})$$

by Lemmas 03S6 and 03S7. Thus it suffices to show the lemma for $\pi : Y \to X$ finite étale. This is Lemma 095B. □